Jihwan Oh

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RESEARCH INTERESTS

- Cross-layer optimization and hardware–software co-design for energy-efficient, high-performance computing
- Computer Architecture, Specialized Accelerators, and Systems for ML

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Feb. 2019 – Feb. 2026
  B.S. in School of Electrical Engineering (Primary Major), School of Computing (Double Major)
  GPA: 4.07/4.3, *Summa Cum Laude (expected)*; Dean's List: Spring 2019, Spring 2024
- **Georgia Institute of Technology** Jan. 2025 – Jul. 2025
  Exchange Program, School of Electrical and Computer Engineering; GPA: 4.0/4.0
- **Chungbuk Science High School** Mar. 2017 – Feb. 2019
  Selected for Early Graduation to KAIST

PUBLICATIONS

Characterizing Compute–Communication Overlap in GPU-Accelerated Distributed Deep Learning: Performance and Power Implications May. 2025
Seonho Lee, ***Jihwan Oh***, Junkyum Kim, Seokjin Go, Jongse Park, Divya Mahajan
Poster presented at ISPASS 2025, arXiv:2507.03114

Mitigating Compute-Communication Overlap Overheads in NCCL Using TMA-Driven Shared-Memory Staging (In progress)
Jihwan Oh, Seokjin Go, Junkyum Kim, Jongse Park, Divya Mahajan

RESEARCH EXPERIENCE

Systems Infrastructure and Architecture Research Lab, Georgia Tech Jan. 2025 – Present
Undergraduate Researcher (Advisor: Prof. Divya Mahajan)
- Characterized compute–communication overlap in large-scale distributed LLM training and showed that the widely held assumption "maximizing overlap is always beneficial" breaks down, with aggressive overlap degrading both execution time and power efficiency across parallelism configurations.
- Led a project to identify the root causes of compute–communication overlap overheads hidden at the software level through hardware-level analysis.
- Designed a new NCCL cross-layer communication protocol that repurposes unused shared memory as a TMA-driven staging buffer, achieving comparable or higher bandwidth with 50% fewer SMs and currently integrating it into NCCL.

Computer Architecture and Systems Lab, KAIST Sep. 2024 – Dec. 2024
Individual Researcher (Advisor: Prof. Jongse Park)
- Analyzed LLM inference on *NeuPIMs* (ASPLOS 2024) and found that FFN compute grows quadratically with hidden size, outpacing other operations and causing severe NPU–PIM load imbalance that limits model-size scalability. Designed a redistribution strategy that balances FFN execution across NPU and PIM, improving utilization and end-to-end performance.
- Participated in research optimizing quantization operations on Processing-in-Memory (PIM) architectures to improve computational efficiency.

Urban Robotics Lab, KAIST May. 2024 – Jun. 2024
Individual Researcher (Advisor: Prof. Hyun Myung)
- Implemented SLAM algorithms and ROS-based path planning systems for mobile robotics.

TEACHING & INDUSTRY EXPERIENCE

CS.10001: Introduction to Programming, KAIST, Teaching Assistant — Sep. 2025 – Present

Yulmae Academy, Programming Instructor — Dec. 2023 – Dec. 2024

- Taught C, Python, and introductory machine learning to science high school students.

CS.93000: Immersion Camp (Coding Camp), KAIST, Teaching Assistant — Aug. 2023 – Feb. 2024

Republic of Korea Air Force, Software Developer — Aug. 2021 – May. 2023

- Developed and refactored a VR flight-training system using Unreal Engine 4 and C++, improving realism and training efficiency.
- Collaborated with a small team of developers using Git for version control and incorporated feedback from real pilots to refine the simulator.
- Served as a counseling officer, supporting new soldiers' adaptation, managing dormitory operations, and assisting with mental-health and discipline issues.

ACADEMIC ACTIVITIES

Poster Presentation, ISPASS 2025 – Ghent, Belgium — May. 2025

- Presented poster on "Characterizing Compute–Communication Overlap in GPU-Accelerated Distributed Deep Learning".

Attendee, ISCA 2025 and uArch Workshop – Tokyo, Japan — Jun. 2025

- Selected as a full travel grant recipient to attend ISCA 2025 and participate in the uArch Workshop.

AWARDS & SCHOLARSHIPS

- **Next-Generation Engineer Award: Highest Distinction** — 2025
  (Institute for Promotion of Engineering and Science of Korea, IPESK)
  Awarded the top distinction as the single highest-ranked recipient in the department.
- **Undergraduate Architecture Mentoring Workshop Travel Grant** — 2025
  7th uArch Workshop / ISCA 2025
- **Student Travel Grant**, IEEE ISPASS 2025 — 2025
- **Dean's List**, School of Electrical Engineering, KAIST — 2024
- **Korea–U.S. Student Exchange Program Scholarship**,
  (Korea Institute for Advancement of Technology, KIAT) — 2024
- **National Science and Technology Scholarship**, Korea Government — 2022 – 2024
- **Dean's List**, Freshman Division, KAIST — 2019
- **Minister of Education Award**, Ministry of Education (STEAM R&E Program) — 2018

EXTRACURRICULAR ACTIVITIES

KAIST Buddy Program — 2025

- Supported incoming international students by facilitating academic and cultural onboarding

ICISTS, Public Relations Head — 2019 – 2021

- Designed and executed PR campaigns for the student-led startup conference *GRAFFITI*
- Coordinated founders' talks and project-based sessions, bridging industry and academia
- Grew attendance to 150+ participants through targeted outreach and branding

Summer Mentoring Camp, Rural Education Initiative — 2019

- Delivered on-site STEM lessons and career mentoring sessions for middle school students in rural Korea

Technical Skills

- **Programming Languages:** C/C++, Python, Rust, SystemVerilog, Verilog
- **Libraries and Tools:** CUDA, PyTorch, NCCL, CUTLASS, cuBLAS; Megatron-LM, Megatron-DeepSpeed; Nsight Compute, Nsight Systems; Unreal Engine 4; Git